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Winter, 2026

B. Tech 4th Semester Examination

Construction Engineering and Management- Theory

Course Code: BCE23406T

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

PART-A

1. Choose the correct answer. **(1 x 10 = 10)**

i. The first stage of construction is
(a) Preparation of estimate, (b) Survey of the site, (c) Initiation of proposal, (d) Preparation of Tender

ii. For the execution of a project, a contractor is
(a) An individual, (b) A firm, (c) An agency, (d) All the above

iii. Which chart is commonly used for project scheduling?
(a) Pie chart, (b) Histogram, (c) Gantt chart, (d) Scatter plot

iv. Detailed construction planning mainly focuses on:
(a) Site execution details and scheduling, (b) Demolition process
(c) Advertising the project, (d) Labour strikes

v. Formwork is mainly used for:
(a) Soil compaction, (b) Supporting fresh concrete till hardening
(c) Brick transportation, (d) Dewatering

vi. Which foundation is generally suitable for weak soil conditions?
(a) Shallow foundation, (b) Raft foundation, (c) Strip footing, (d) Isolated footing

vii. Which of the following is used to identify project completion probability?
(a) CPM, (b) PERT, (c) Bar chart, (d) Histogram

viii. In PERT analysis, how many time estimates are generally considered?
(a) One, (b) Two, (c) Three, (d) Four

ix. In CPM, the critical path is the path having:
(a) Maximum float, (b) Minimum activities, (c) Zero float, (d) Maximum cost

x. The major advantage of mechanized construction over conventional construction is:
(a) Increased labour dependency, (b) Lower speed of execution
(c) Improved productivity and faster work, (d) Reduced accuracy

PART-B

2. The following table lists activities of a project along with their duration in days. Construct the network and compute in tabular form their early start, early finish, late start and late finish times. Determine the critical path and find the total float, free float and independent float for all the activities. **(7)**

Activity	Duration (days)
1-2	20
1-3	25
2-3	10
2-4	12
3-4	6
4-5	10

3. (a) Differentiate between CPM and PERT. (3)

Or,

Discuss various stages involved in the construction of a project.

(b) The network for a certain project is given as follows along with their estimated times. (6)

i) Find its critical path and expected project duration.

ii) Calculate the variances of each of the activities.

iii) What is the probability of completion of the project if the scheduled duration is 31 days.

Activity	t_o	t_m	t_p
1-2	6	8	11
1-3	3	7	9
1-4	5	7	10
2-7	8	10	12
2-6	4	8	12
3-6	8	10	12
4-5	4	6	8
5-6	3	5	6
7-8	5	8	12
6-8	7	10	15

4. Write a short note on types of foundations used in construction. (4)

Or,

Discuss the advantages of mechanized construction

5. Discuss the common good practices in construction. (5)

Or,

What is Cost Over run and Time Over run? Discuss their corrective measures in details.

7. Discuss in details the basics of Modern Project Management Systems. (3)
8. Define contract system. Discuss in details the types of contract agreement. (6)

Or,

Differentiate between Security Deposit and Retention Money. Name (any six) common contract clauses.

9. Discuss in details the direct and indirect construction costs. (6)

Or,

Discuss time-cost trade-off in details.